

8.7 Affine Transformation



Core skills you will master in this section:

- ▶ Distinguish linear transformation from affine transformation: a linear transformation leaves the origin fixed, while an affine transformation includes translation.
- ▶ Introduce homogeneous coordinates: translation, along with every geometric operation covered earlier in this chapter, can be unified as matrix multiplication.
- ▶ Use 3×3 matrix multiplication to carry out planar affine transformations.
- ▶ Use 4×4 matrix multiplication to carry out affine transformations in 3D space.
- ▶ Use block matrix multiplication to understand affine-transformation matrix multiplication.

As mentioned earlier, a linear transformation is a special case of an affine transformation (affine transformation). The difference between the two is that affine transformation includes translation.

A linear transformation can only scale, rotate, shear, and so on — it never moves the origin, meaning the origin is always a fixed point. In other words, under a linear transformation, the origin is always mapped to the origin.

An affine transformation, on the other hand, adds a translation on top of a linear transformation, allowing the shape to move to a new location. In other words, the origin can be moved elsewhere.

This section introduces several common affine transformations; please read it alongside the earlier material in this chapter for comparison.

Planar Translation

When coordinates are expressed as column vectors, translation can be written as:

$$\begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \mathbf{t}$$

where $\mathbf{t}$ is the translation vector

$$\mathbf{t} = \begin{bmatrix} t_1 \\ t_2 \end{bmatrix}$$

Substituting the equation above into the previous equation gives:

$$\begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} t_1 \\ t_2 \end{bmatrix} = \begin{bmatrix} x_1 + t_1 \\ x_2 + t_2 \end{bmatrix}$$

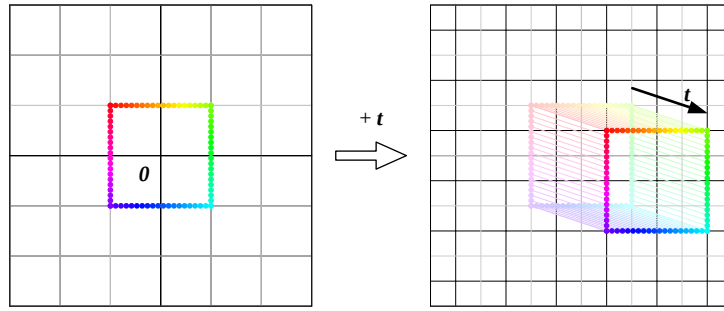


Figure 1. Vector Addition for Planar Translation

From a 2x2 Matrix to a 3x3 Matrix

We use 3×3 matrices to carry out planar affine transformations because the introduction of homogeneous coordinates (homogeneous coordinates) lets us conveniently express geometric transformations (such as translation, scaling, rotation, and so on) via matrix multiplication, while also supporting composite transformations.

Simply put, homogeneous coordinates represent what was originally an n -dimensional vector using an $(n + 1)$ -dimensional vector instead.

Earlier in this chapter we saw that a 2×2 matrix alone cannot describe planar translation. But with a 3×3 matrix, planar geometric transformations – including translation – can all be carried out via the equation below, specifically as follows:

$$\begin{bmatrix} z_1 \\ z_2 \\ 1 \end{bmatrix} = \mathbf{A}_{3 \times 3} \begin{bmatrix} x_1 \\ x_2 \\ 1 \end{bmatrix}$$

This way, for a sequence of geometric transformations, we simply need to multiply their corresponding matrices.

Similarly, 3D geometric transformations can also be carried out via 4×4 matrix multiplication.

Let's begin with planar translation.

Matrix Multiplication for Planar Translation

In the plane, translation moves a shape in a given direction without changing its shape or size.

Using a 3×3 matrix, planar translation can be carried out via the following matrix multiplication

$$\begin{bmatrix} z_1 \\ z_2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_1 \\ 0 & 1 & t_2 \\ 0 & 0 & 1 \end{bmatrix} @ \begin{bmatrix} x_1 \\ x_2 \\ 1 \end{bmatrix}$$

where t_1 represents the shape's horizontal displacement; t_2 represents its vertical displacement.

Expanding the equation above gives two equalities:

$$\begin{cases} z_1 = x_1 + t_1 \\ z_2 = x_2 + t_2 \end{cases}$$

The inverse of the translation above can also be carried out via matrix multiplication:

$$\begin{bmatrix} 1 & 0 & t_1 \\ 0 & 1 & t_2 \\ 0 & 0 & 1 \end{bmatrix}^{-1} @ \begin{bmatrix} z_1 \\ z_2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -t_1 \\ 0 & 1 & -t_2 \\ 0 & 0 & 1 \end{bmatrix} @ \begin{bmatrix} z_1 \\ z_2 \\ 1 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ 1 \end{bmatrix}$$

This equation also tells us that if an upper triangular matrix is invertible, its inverse is also upper triangular. An upper triangular matrix is invertible if and only if all of its diagonal elements are nonzero.

? Figure 2 gives an example; please analyze it on your own.

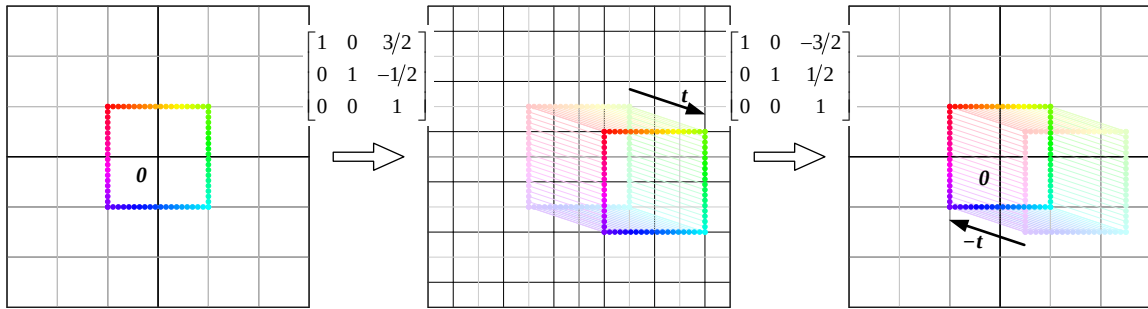


Figure 2. Using 3×3 Matrix Multiplication for Planar Translation and Its Inverse

Block Matrix Multiplication

From the perspective of block matrix multiplication, this operation is actually quite easy to understand. First, partition the 3×3 translation matrix into blocks:

$$\begin{bmatrix} 1 & 0 & t_1 \\ 0 & 1 & t_2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} I_{2 \times 2} & \mathbf{t} \\ \mathbf{0}_{1 \times 2} & 1 \end{bmatrix}$$

This way, the multiplication corresponding to translation can be written as

$$\begin{bmatrix} I_{2 \times 2} & \mathbf{t} \\ \mathbf{0}_{1 \times 2} & 1 \end{bmatrix} @ \begin{bmatrix} \mathbf{x} \\ 1 \end{bmatrix} = \begin{bmatrix} I_{2 \times 2} \mathbf{x} + \mathbf{t} \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{z} \\ 1 \end{bmatrix}$$

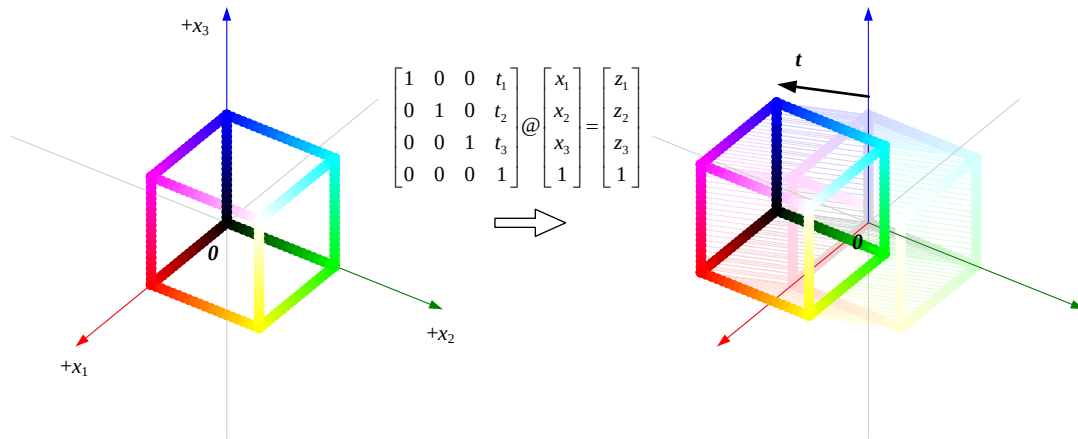
We're sure you've already noticed the addition equality $\mathbf{x} + \mathbf{t} = \mathbf{z}$.

Translation in 3D Space

Figure 3 shows translation in 3D space carried out via 4×4 matrix multiplication; the corresponding matrix multiplication is

$$\begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & t_1 \\ 0 & 1 & 0 & t_2 \\ 0 & 0 & 1 & t_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} @ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ 1 \end{bmatrix}$$

? Please work out on your own, in Figure 3, the matrix multiplication corresponding to the reverse translation.

Figure 3. Using 4×4 Matrix Multiplication for Translation in 3D Space

Planar Scaling

Recall the planar scaling introduced earlier in this chapter: uniform planar scaling keeps a shape's proportions consistent and changes only its size.

Using a 3×3 matrix, planar uniform scaling can be carried out via the following multiplication

$$\begin{bmatrix} z_1 \\ z_2 \\ 1 \end{bmatrix} = \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & 1 \end{bmatrix} @ \begin{bmatrix} x_1 \\ x_2 \\ 1 \end{bmatrix}$$

where s is the scaling factor. $s > 1$ means enlargement; $0 < s < 1$ means shrinking; $s = 1$ means no scaling. When $s < 0$, in addition to scaling there is also a reflection, equivalent to a composite geometric transformation. In the special case $s = 0$, this becomes a projection that reduces dimensionality.



Please also use block matrix multiplication to expand the equation above.

Using the equation above, the inverse of uniform scaling is

$$\begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} @ \begin{bmatrix} z_1 \\ z_2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/s & 0 & 0 \\ 0 & 1/s & 0 \\ 0 & 0 & 1 \end{bmatrix} @ \begin{bmatrix} z_1 \\ z_2 \\ 1 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ 1 \end{bmatrix}$$

Non-uniform scaling has different scaling ratios in different directions.

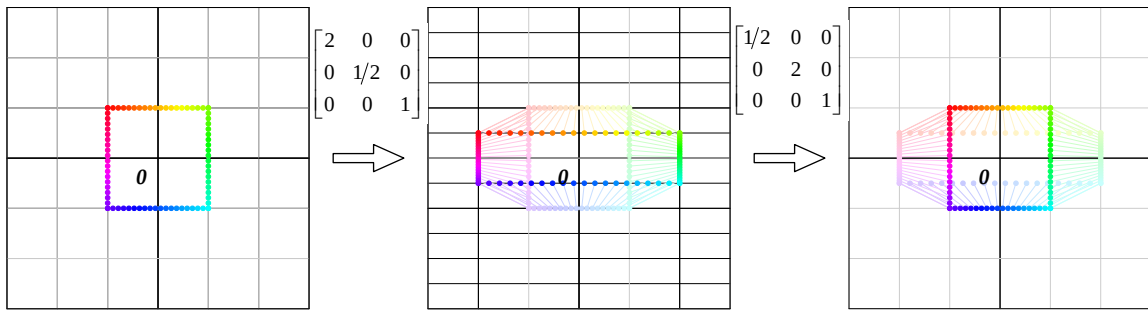
Using a 3×3 matrix, planar non-uniform scaling can be carried out via the following multiplication

$$\begin{bmatrix} z_1 \\ z_2 \\ 1 \end{bmatrix} = \begin{bmatrix} s_1 & 0 & 0 \\ 0 & s_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} @ \begin{bmatrix} x_1 \\ x_2 \\ 1 \end{bmatrix}$$

where s_1 is the horizontal scaling factor; s_2 is the vertical scaling factor.



Please analyze Figure 4, and please also use block matrix multiplication to expand the equation above.

Figure 4. Using 3×3 Matrix Multiplication for Non-Uniform Scaling and Its Inverse

Planar Rotation

As mentioned earlier in this chapter, a planar (about the origin) counterclockwise rotation transformation rotates a shape counterclockwise about the origin by some angle.

Using a 3×3 matrix, planar (about the origin) counterclockwise rotation can be carried out via the following multiplication

$$\begin{bmatrix} z_1 \\ z_2 \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} @ \begin{bmatrix} x_1 \\ x_2 \\ 1 \end{bmatrix}$$

where θ is the rotation angle, with counterclockwise as the positive direction.



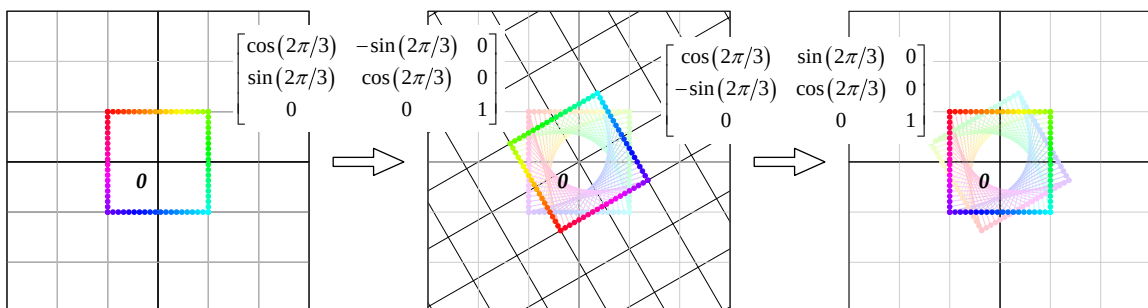
Please also use block matrix multiplication to expand the equation above.

Using the equation above, the inverse of the rotation is

$$\begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} @ \begin{bmatrix} z_1 \\ z_2 \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} @ \begin{bmatrix} z_1 \\ z_2 \\ 1 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ 1 \end{bmatrix}$$



Please analyze the matrix multiplication corresponding to Figure 5.

Figure 5. Using 3×3 Matrix Multiplication for Rotation and Its Inverse

Composition

Using 3×3 matrices, planar geometric operations such as translation, scaling, and rotation can be carried out in different orders via matrix multiplication, which significantly improves the flexibility and computational efficiency of geometric transformations.

Through the associativity of matrix multiplication, multiple geometric operations can be combined into a single matrix, simplifying the execution of a transformation to a single matrix multiplication rather than requiring separate operations on each point step by step. This approach is of great significance for geometric computation, image processing, and computer graphics.

? It's worth emphasizing repeatedly that, in general, matrix multiplication is not commutative — this means that different orders of operations produce different results.

Figure 6 shows translation followed by rotation. The shape is moved first, then rotated about the origin; the amount of translation is affected by the rotation angle.

Figure 6 corresponding matrix multiplication is shown below

$$\begin{bmatrix} z_1 \\ z_2 \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\text{2nd, rotate}} @ \underbrace{\begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}}_{\text{1st, translate}} = \begin{bmatrix} \cos \theta & -\sin \theta & \cos \theta \cdot t_x - \sin \theta \cdot t_y \\ \sin \theta & \cos \theta & \sin \theta \cdot t_x + \cos \theta \cdot t_y \\ 0 & 0 & 1 \end{bmatrix} @ \begin{bmatrix} x_1 \\ x_2 \\ 1 \end{bmatrix}$$

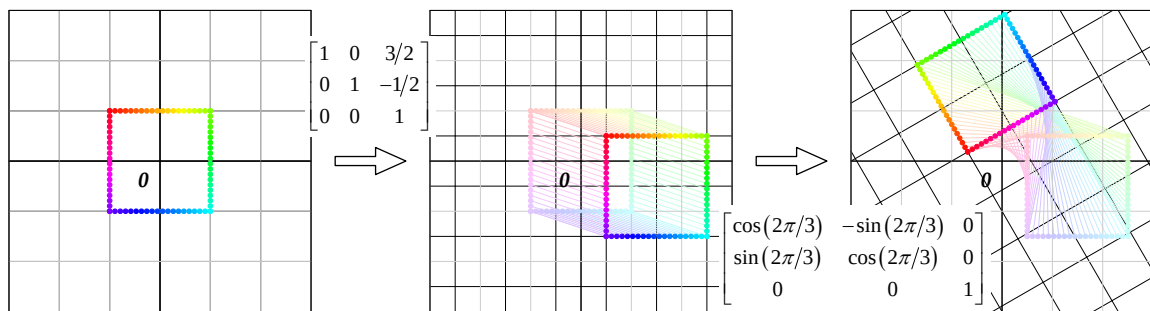
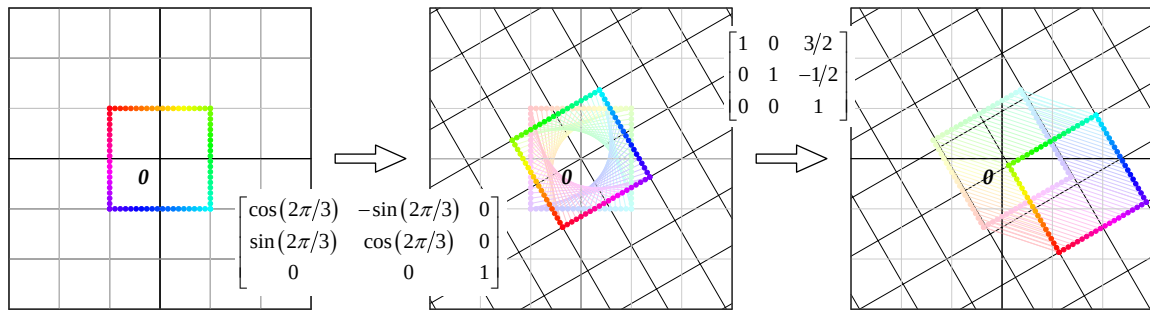


Figure 6. Using 3×3 Matrix Multiplication for Translation Followed by Rotation

Figure 7 shows rotation followed by translation. The shape is rotated about the origin first, then moved; the translation is applied directly along the global coordinate axes and is unaffected by the rotation.

Figure 7 corresponding matrix multiplication is shown below

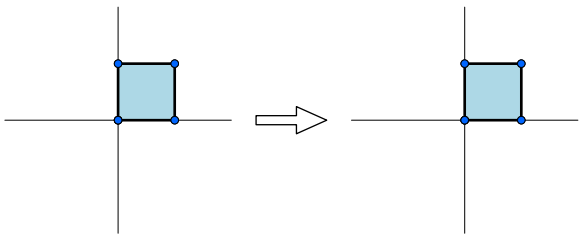
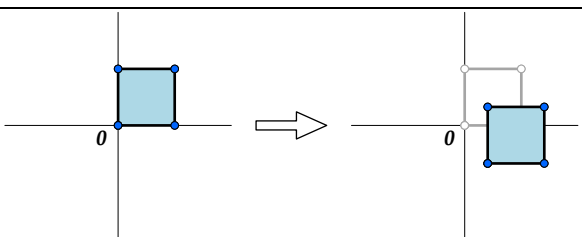
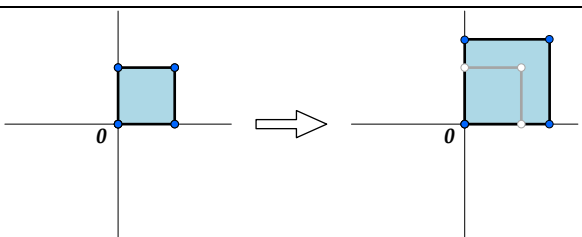
$$\begin{bmatrix} z_1 \\ z_2 \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}}_{\text{2nd, translate}} @ \underbrace{\begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\text{1st, rotate}} = \begin{bmatrix} \cos \theta & -\sin \theta & t_x \\ \sin \theta & \cos \theta & t_y \\ 0 & 0 & 1 \end{bmatrix} @ \begin{bmatrix} x_1 \\ x_2 \\ 1 \end{bmatrix}$$

Figure 7. Using 3×3 Matrix Multiplication for Rotation Followed by Translation

An affine transformation can be viewed as an extended form of a linear transformation with translation added, giving it a broader scope of application, while a linear transformation can be viewed as a special case of an affine transformation. Through this section, we hope you've seen that because it includes translation, affine transformation makes it convenient, efficient, and flexible to carry out all kinds of shape transformations, and is therefore more widely used in computer graphics and image processing.

 Table 1 compares the planar linear-transformation and affine-transformation matrices; please analyze them one by one.

Table 1. Comparing Linear Transformation and Affine Transformation in the Plane

Linear Transformation	Affine Transformation	Geometric Transformation
No transformation (identity matrix) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ Determinant 1	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	
Translation Not a linear transformation	$\begin{bmatrix} 1 & 0 & t_1 \\ 0 & 1 & t_2 \\ 0 & 0 & 1 \end{bmatrix}$	
Uniform scaling $\begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix}$ Determinant s^2	$\begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & 1 \end{bmatrix}$	

Non-uniform scaling $\begin{bmatrix} s_1 & 0 \\ 0 & s_2 \end{bmatrix}$ Determinant $s_1 s_2$	$\begin{bmatrix} s_1 & 0 & 0 \\ 0 & s_2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	
Rotation $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ Determinant 1	$\begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$	
Shear along the x_1 axis $\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$ Determinant 1	$\begin{bmatrix} 1 & k & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	
Shear along the x_2 axis $\begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$ Determinant 1	$\begin{bmatrix} 1 & 0 & 0 \\ k & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	
Orthogonal projection $\frac{1}{a_1^2 + a_2^2} \begin{bmatrix} a_2^2 & -a_1 a_2 \\ -a_1 a_2 & a_1^2 \end{bmatrix}$ (through the origin) line normal vector $\mathbf{n} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ Determinant 0	$\begin{bmatrix} \frac{a_2^2}{a_1^2 + a_2^2} & \frac{-a_1 a_2}{a_1^2 + a_2^2} & 0 \\ \frac{-a_1 a_2}{a_1^2 + a_2^2} & \frac{a_1^2}{a_1^2 + a_2^2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$	
Projection onto the x_1 axis $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ Determinant 0	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	
Projection onto the x_2 axis $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ Determinant 0	$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	

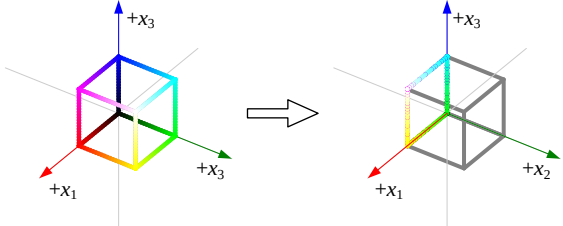
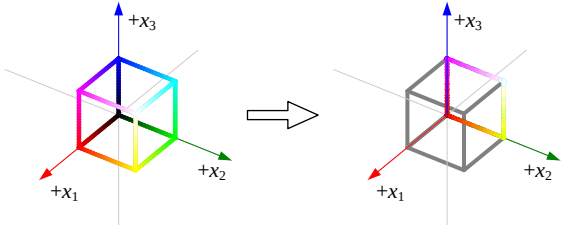
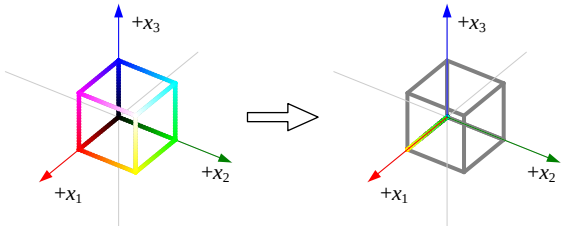
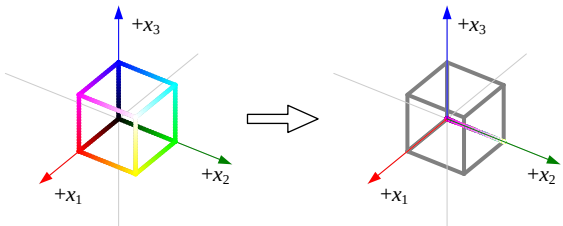
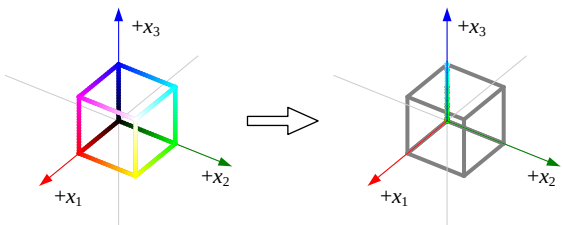
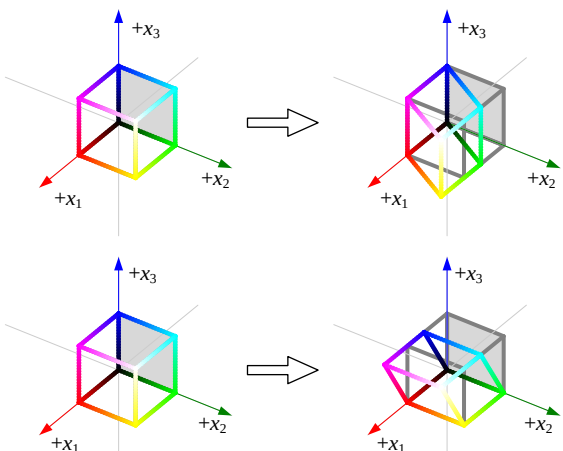
Reflection $\frac{1}{a_1^2 + a_2^2} \begin{bmatrix} a_2^2 - a_1^2 & -2a_1a_2 \\ -2a_1a_2 & a_1^2 - a_2^2 \end{bmatrix}$ (through the origin) line normal vector $\mathbf{n} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ Determinant -1	$\begin{bmatrix} \frac{a_2^2 - a_1^2}{a_1^2 + a_2^2} & \frac{-2a_1a_2}{a_1^2 + a_2^2} & 0 \\ \frac{-2a_1a_2}{a_1^2 + a_2^2} & \frac{a_1^2 - a_2^2}{a_1^2 + a_2^2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$	
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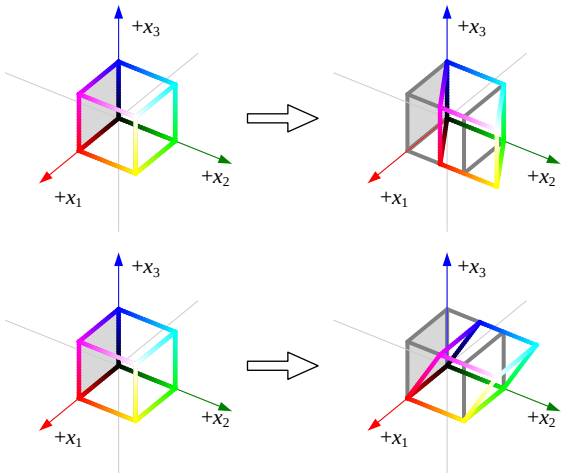
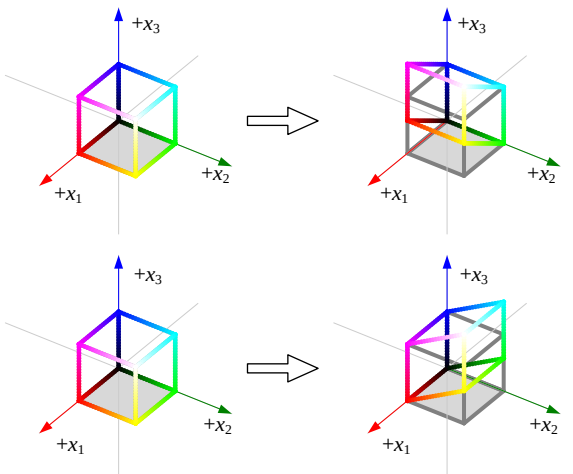
? Table 2 compares the linear-transformation and affine-transformation matrices in 3D space; please analyze them one by one.

Table 2. Comparing Linear Transformation and Affine Transformation in 3D Space

Linear Transformation	Affine Transformation	Example
No transformation (identity matrix) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ Determinant 1	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	
Translation Not a linear transformation	$\begin{bmatrix} 1 & 0 & 0 & t_1 \\ 0 & 1 & 0 & t_2 \\ 0 & 0 & 1 & t_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	
Uniform scaling by factor s $\begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix}$ Determinant s^3	$\begin{bmatrix} s & 0 & 0 & 0 \\ 0 & s & 0 & 0 \\ 0 & 0 & s & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	
Non-uniform scaling $\begin{bmatrix} s_1 & 0 & 0 \\ 0 & s_2 & 0 \\ 0 & 0 & s_3 \end{bmatrix}$ Determinant $s_1s_2s_3$	$\begin{bmatrix} s_1 & 0 & 0 & 0 \\ 0 & s_2 & 0 & 0 \\ 0 & 0 & s_3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	

Counterclockwise rotation about the x_1 axis $\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta_1 & -\sin \theta_1 \\ 0 & \sin \theta_1 & \cos \theta_1 \end{bmatrix}$ Determinant 1	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta_1 & -\sin \theta_1 & 0 \\ 0 & -\sin \theta_1 & \cos \theta_1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	
Counterclockwise rotation about the x_2 axis $\begin{bmatrix} \cos \theta_2 & 0 & \sin \theta_2 \\ 0 & 1 & 0 \\ -\sin \theta_2 & 0 & \cos \theta_2 \end{bmatrix}$ Determinant 1	$\begin{bmatrix} \cos \theta_2 & 0 & \sin \theta_2 & 0 \\ 0 & 1 & 0 & 0 \\ -\sin \theta_2 & 0 & \cos \theta_2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	
Counterclockwise rotation about the x_3 axis $\begin{bmatrix} \cos \theta_3 & -\sin \theta_3 & 0 \\ \sin \theta_3 & \cos \theta_3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ Determinant 1	$\begin{bmatrix} \cos \theta_3 & -\sin \theta_3 & 0 & 0 \\ \sin \theta_3 & \cos \theta_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	
Reflection about the x_1x_2 plane $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ Determinant -1	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	
Reflection about the x_1x_3 plane $\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ Determinant -1	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	
Reflection about the x_2x_3 plane $\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ Determinant -1	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	
Projection onto the x_1x_2 plane $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ Determinant 0	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	

Projection onto the x_1x_3 plane $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ Determinant 0	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	
Projection onto the x_2x_3 plane $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ Determinant 0	$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	
Projection onto the x_1 axis $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ Determinant 0	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	
Projection onto the x_2 axis $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ Determinant 0	$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	
Projection onto the x_3 axis $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ Determinant 0	$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	
Shear along the x_1 axis $\begin{bmatrix} 1 & k_{1,2} & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & k_{1,3} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ Determinant 1	$\begin{bmatrix} 1 & k_{1,2} & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & k_{1,3} & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	

Shear along the x_2 axis $\begin{bmatrix} 1 & 0 & 0 \\ k_{2,1} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & k_{2,3} \\ 0 & 0 & 1 \end{bmatrix}$ Determinant 1	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ k_{2,1} & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & k_{2,3} & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	
Shear along the x_3 axis $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ k_{3,1} & 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & k_{3,2} & 1 \end{bmatrix}$ Determinant 1	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ k_{3,1} & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & k_{3,2} & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. What is the essential difference between a linear transformation and an affine transformation?

Q2. Why is translation not a linear transformation?

Q3. In the plane, vector x is translated by $[1, 1]^T$ — how can this operation be carried out via matrix multiplication?

Q4. To undo the translation in Q3, how would you carry out the operation using vector addition/subtraction and matrix multiplication, respectively?

Q5. How would you use matrix multiplication to carry out, in the plane: 1) first a counterclockwise rotation of 90° about the origin; 2) then a translation by $[1, 1]^T$?

Q6. How would you use matrix multiplication to carry out, in the plane: 1) first a translation by $[1, 1]^T$; 2) then a counterclockwise rotation of 90° about the origin?

Q7. How would you use matrix multiplication to carry out, in 3D space: 1) first a translation by $[1, 1, 1]^T$; 2) then a counterclockwise rotation about the x_1 axis by 90° ?